Generalised Challenge 20

Two arbitrarily shaped particles, the multipole hierarchy, and why the two-ellipsoid answer survives

Companion code: `generalized.py` · reference solution: `../answer.py`

Short answer

Yes. Solve the general problem once — two rigid dielectric bodies of *arbitrary* shape in the same tweezer geometry, each described by its full multipole (T-matrix) response — and the two-ellipsoid result drops out as the leading term of a controlled double expansion in a/λ (multipole order of the single-particle response) and a/R (multipole order of the interaction). The reduction is sharper than a mere limit:

1. At dipole order the physics depends on the shape *only* through the symmetric rank-2 polarisability tensor $\boldsymbol{\alpha}$ and the inertia tensor I. Since every admissible $\boldsymbol{\alpha}$ is realised by exactly one ellipsoid (Sec. 2.1, closed form), **an arbitrary sub-wavelength particle is optically indistinguishable from its "equivalent ellipsoid".** Only I has to be carried over separately.
2. What the general problem *adds* is structure the challenge's Hamiltonian hides: two librational modes per particle instead of one (four modes, generally non-degenerate), a coupling *matrix* that samples the longitudinal Green function $G_\parallel \sim 1/R^2$ as well as the transverse $G_\perp \sim 1/R$, an exact zero mode (spin about the polarisation), and — for shapes without inversion symmetry — an $O(a/R)$ dipole–quadrupole correction that *no* ellipsoid can reproduce.
3. Set the particles axially symmetric, put the polarisation along $\mathbf{R}$, and keep one libration plane: the four-mode problem collapses to $H = \hbar\omega_t \sum_i a_i^\dagger a_i + \hbar g(a_1^\dagger a_2 + \text{h.c.})$ with exactly the ω_t and g of `answer.py`. Verified numerically to 9×10^{-16} relative (Sec. 4).

1 The general problem

Two rigid dielectric bodies of arbitrary shape, permittivity ϵ_r and density ρ, sit at positions $\mathbf{r}_{1,2}$ with orientations $\Omega_{1,2} \in SO(3)$, separated by $\mathbf{R} = R\hat{R}$, each trapped by a linearly polarised Gaussian tweezer of amplitude E_0, polarisation $\hat{e}$ and wave vector k. The exact conservative light–matter energy is obtained by solving the two-body scattering problem self-consistently; writing the response of body i as its T-matrix $\mathsf{T}_i(\Omega_i)$ and the free-space propagator as $\mathsf{G}(\mathbf{R})$,

$$U = -\tfrac{1}{2}\,\mathrm{Re}\Big[\mathbf{E}_0^\dagger \mathsf{T}_1 \mathbf{E}_0 + \mathbf{E}_0^\dagger \mathsf{T}_2 \mathbf{E}_0 + 2\,\mathbf{E}_0^\dagger \mathsf{T}_1 \mathsf{G} \mathsf{T}_2 \mathbf{E}_0 + O(\mathsf{T}^3)\Big], \tag{1}$$

the third term being the optical binding energy at lowest order in multiple scattering. In the quasi-static (Rayleigh) regime it is more convenient to expand T in Cartesian multipole polarisabilities,

$$\mathbf{p} = \boldsymbol{\alpha}\cdot\mathbf{E} + \tfrac{1}{3}\mathsf{A}\!:\!\nabla\mathbf{E} + \dots, \qquad \mathsf{Q} = \mathsf{A}\cdot\mathbf{E} + \mathsf{C}\!:\!\nabla\mathbf{E} + \dots, \tag{2}$$

with $\boldsymbol{\alpha}$ rank 2, A rank 3 and C rank 4, and to write the interaction as

$$U_{\text{int}} = -C\,\mathrm{Re}\Big[\mathbf{p}_1\cdot\mathbf{E}_2 + \tfrac{1}{3}\mathsf{Q}_1\!:\!\nabla\mathbf{E}_2 + \dots\Big], \qquad C = \tfrac{1}{2}\ \text{(cycle average)}. \tag{3}$$

(The value of C is the factor-of-two issue flagged in `../challenge20.pdf`; it propagates unchanged into everything below.)

Power counting. With a the particle size, $\mathbf{p} \sim \epsilon_0 a^3 E$, $\nabla \mathbf{E}_2 \sim E_2/R$ and $\mathsf{Q} \sim \epsilon_0 a^4 E$ (if $\mathsf{A} \neq 0$) or $\epsilon_0 a^5 \nabla E$ (if $\mathsf{A} = 0$). Hence

$$\frac{U_{\rm int}}{U_{\rm int}^{\rm dip-dip}} = 1 + \underbrace{O\Big(\frac{a}{R}\Big)}_{\text{only if } \mathsf{A}\neq 0} + O\Big(\frac{a}{R}\Big)^2 + \dots, \qquad \frac{\boldsymbol{\alpha}_{\rm exact}}{\boldsymbol{\alpha}_{\rm static}} = 1 + O\Big(\frac{a}{\lambda}\Big)^2. \tag{4}$$

A **vanishes identically for any centrosymmetric body**, an ellipsoid included, because A is odd under $\mathbf{r} \to -\mathbf{r}$. So for ellipsoids the series in a/R is even — which is why Appendix A of Ref. [1] finds a leading correction $\propto (a^2 - b^2)/R^2$ even though it calls it "first order". *The odd term is the one genuinely new effect an arbitrary shape brings to the interaction.*

2 Dipole order in full generality

2.1 The equivalent ellipsoid

At leading order every sub-wavelength body is a point dipole with a symmetric positive-definite $\boldsymbol{\alpha}$; let $\alpha_1 \geq \alpha_2 \geq \alpha_3$ be its principal values. For an ellipsoid with semi-axes (a, b, c),

$$\alpha_i = \frac{\epsilon_0 V(\epsilon_r - 1)}{1 + L_i(\epsilon_r - 1)}, \qquad V = \tfrac{4\pi}{3} abc, \qquad L_i = \frac{abc}{2} \int_0^\infty \frac{\mathrm{d}s}{(s + a_i^2)\sqrt{(s+a^2)(s+b^2)(s+c^2)}}, \tag{5}$$

with $\sum_i L_i = 1$ [7, 8]. Inverting Eq. (5) and imposing the sum rule fixes the volume in *closed form*,

$$\boxed{V = \frac{\epsilon_r + 2}{\epsilon_0(\epsilon_r - 1)\sum_i \alpha_i^{-1}}}, \qquad L_i = \frac{1}{\epsilon_r - 1}\left[\frac{\epsilon_0 V(\epsilon_r - 1)}{\alpha_i} - 1\right], \tag{6}$$

after which the two independent axis ratios follow from the L_i by a two-dimensional root find (the map from shape to the open simplex $\{L_i > 0, \sum L_i = 1\}$ is a bijection, so the ellipsoid is unique up to relabelling). `generalized.py` implements this and round-trips it to 10^{-11}.

> **Equivalent-ellipsoid statement.** At dipole order the two-particle librational problem depends on the shape only through $\boldsymbol{\alpha}$ and I. Equation (6) produces the unique ellipsoid with the same $\boldsymbol{\alpha}$; therefore *the optical half of the arbitrary-shape problem is exactly the ellipsoid problem.* The mechanical half is not: the equivalent ellipsoid generally has the wrong I, and for a low-symmetry body the principal axes of $\boldsymbol{\alpha}$ and of I need not even coincide.

2.2 Equilibrium, the zero mode, and two librations per particle

The orientational potential of one particle in a uniform linearly polarised field is $U(\Omega) = -\frac{1}{4}E_0^2\, \hat{e} \cdot \boldsymbol{\alpha}(\Omega) \cdot \hat{e}$. It is stationary when $\hat{e}$ is an eigenvector of $\boldsymbol{\alpha}$ and minimal when $\hat{e}$ lies along the most polarisable body axis $\hat{u}_1$. Parametrising a small rotation by the vector $\boldsymbol{\varphi}$ about equilibrium,

$$U = \text{const} + \tfrac{1}{4}E_0^2\left[(\alpha_1 - \alpha_2)\varphi_3^2 + (\alpha_1 - \alpha_3)\varphi_2^2\right], \qquad k_2 = \tfrac{1}{2}E_0^2(\alpha_1 - \alpha_3),\ k_3 = \tfrac{1}{2}E_0^2(\alpha_1 - \alpha_2). \tag{7}$$

> **Exact zero mode — true for *any* shape**
>
> φ_1, the spin about the polarisation direction, does not appear in Eq. (7): rotating the body about $\hat{e}$ leaves $\hat{e} \cdot \boldsymbol{\alpha} \cdot \hat{e}$ invariant. A uniform, strictly linearly polarised field therefore confines only *two* of the three orientational degrees of freedom, no matter how anisotropic the particle. Confining the third requires the ellipticity and longitudinal field components of a tightly focused beam, a field gradient acting on C, or the neighbour's field — all of which are higher order in the expansion (4).

The two librational frequencies are $\omega_{i\mu} = \sqrt{k_{i\mu}/I_{i\mu}}$, degenerate only when $\alpha_2 = \alpha_3$ and $I_2 = I_3$, i.e. for an axially symmetric particle.

2.3 The coupling matrix

To first order in the tilts the induced dipole in the lab frame is

$$\mathbf{p}_i = E_0\Big[\alpha_1\hat{u}_1 + (\alpha_1-\alpha_2)\varphi_3\,\hat{u}_2 - (\alpha_1-\alpha_3)\varphi_2\,\hat{u}_3\Big] + O(\varphi^2), \tag{8}$$

so that

$$\frac{\partial\mathbf{p}}{\partial\varphi_3} = (\alpha_1-\alpha_2)E_0\hat{u}_2, \qquad \frac{\partial\mathbf{p}}{\partial\varphi_2} = -(\alpha_1-\alpha_3)E_0\hat{u}_3. \tag{9}$$

Inserting into Eq. (3) with $\mathbf{E}_2 = \mathsf{G}(\mathbf{R})\mathbf{p}_2$ gives the whole dipole-order coupling in one line:

$$\boxed{\kappa_{\mu\nu} = -C\,\mathrm{Re}\left[\frac{\partial\mathbf{p}_1}{\partial\varphi^\mu}\cdot\mathsf{G}(\mathbf{R})\cdot\frac{\partial\mathbf{p}_2}{\partial\varphi^\nu}\right]}, \tag{10}$$

$$\mathsf{G}(\mathbf{R}) = \frac{\mathrm{e}^{\mathrm{i}kR}}{4\pi\epsilon_0 R}\left[\left(\nVdash - \hat{R}\hat{R}\right)k^2 + \left(3\hat{R}\hat{R} - \nVdash\right)\left(\frac{1}{R^2} - \frac{\mathrm{i}k}{R}\right)\right]. \tag{11}$$

The eigenvalues of G are

$$G_\parallel = \frac{2\mathrm{e}^{\mathrm{i}kR}(1-\mathrm{i}kR)}{4\pi\epsilon_0 R^3} \xrightarrow{kR\gg1} \frac{2k\sin kR}{4\pi\epsilon_0 R^2}, \tag{12}$$

$$G_\perp = \frac{\mathrm{e}^{\mathrm{i}kR}(k^2R^2+\mathrm{i}kR-1)}{4\pi\epsilon_0 R^3} \xrightarrow{kR\gg1} \frac{k^2\cos kR}{4\pi\epsilon_0 R}, \tag{13}$$

so the longitudinal channel is short ranged ($1/R^2$) and the transverse one long ranged ($1/R$).

2.4 Second quantisation

With $\varphi_{i\mu} = \varphi_{\mathrm{zpf}}^{i\mu}(b_{i\mu} + b_{i\mu}^\dagger)$, $\varphi_{\mathrm{zpf}}^{i\mu} = \sqrt{\hbar/2I_{i\mu}\omega_{i\mu}}$, and the RWA,

$$H = \sum_{i=1,2}\sum_{\mu=2,3}\hbar\omega_{i\mu}b_{i\mu}^\dagger b_{i\mu} + \sum_{\mu\nu}\hbar g_{\mu\nu}\left(b_{1\mu}^\dagger b_{2\nu} + \mathrm{h.c.}\right), \qquad g_{\mu\nu} = \frac{\kappa_{\mu\nu}}{2\sqrt{I_{1\mu}I_{2\nu}\,\omega_{1\mu}\omega_{2\nu}}}. \tag{14}$$

This is the general answer. Everything specific to the challenge is now a matter of specialising $\boldsymbol{\alpha}$, I, $\hat{e}$ and $\hat{R}$.

3 The reduction chain

Step	Assumption	What collapses	Result
0	arbitrary shape, full T	—	Eq. (1)
1	$a \ll \lambda$, $a \ll R$	$\mathsf{A}, \mathsf{C}, \cdots \to 0$	dipole order, Eq. (14)
2	optical response only	shape $\to \boldsymbol{\alpha}$	equivalent ellipsoid, Eq. (6)
3	$\alpha_2 = \alpha_3$, $I_2 = I_3$ (axial symmetry)	$\omega_{i2} = \omega_{i3} \equiv \omega_t$	degenerate librations
4	prolate spheroid	$L_\perp = (1-L_\parallel)/2$	$\Delta\alpha$, $I = m(a^2+b^2)/5$
5	$\hat{e} \parallel \hat{R}$ (the challenge)	$\kappa \propto \nVdash_2\ \mathrm{Re}\,G_\perp$	two identical 2×2 blocks
6	keep one libration plane	$4 \to 2$ modes	H of the challenge

Table 1: From the general problem to `answer.py`.

Step 5 is the reason the challenge's geometry is unusually clean: with $\hat{e} \parallel \hat{R} \parallel \hat{x}$, *both* tilt directions $\hat{u}_2, \hat{u}_3$ are perpendicular to $\hat{R}$, so $\kappa = -C(\Delta\alpha E_0)^2\,\mathrm{Re}\,G_\perp\nVdash_2$: the coupling is isotropic in the transverse plane, the off-diagonal κ_{23} vanishes, and the four-mode problem is two identical copies of the two-mode problem. In the geometry actually used in Ref. [1] ($\hat{e} \perp \hat{R}$) one tilt direction lies along $\hat{R}$ and the two modes couple through $G_\parallel$ and $G_\perp$ respectively — two different couplings with different ranges (see Sec. 4, item 2).

4 Numerical verification

`generalized.py` builds the 4×4 stiffness and inertia matrices from Eqs. (7) and (10) and diagonalises them. With the benchmark parameters ($a = 300$ nm, $b = 180$ nm, $\epsilon_r = 5.7$, $\rho = 3500\,\mathrm{kg\,m^{-3}}$, $w_0 = 500$ nm, $\lambda = 1064$ nm, $R = 1.06\,\mu$m, $P_0 = 1$ W, $C = 1$):

1. **Reduction.** General framework: $\omega_t/2\pi = 1.344893515$ MHz for *both* librations, $g/2\pi = -100.978493209$ kHz for both diagonal entries, $\kappa_{23} = 0$ exactly. `answer.py`: identical to 4×10^{-16} (ω_t) and 9×10^{-16} (g).
2. **Longitudinal vs. transverse.** Rotating the polarisation to $\hat{y}$ (the geometry of Ref. [1]) splits the two couplings: $g_{\|}/2\pi = -4.490$ kHz (in-plane tilt, $G_{\|}$) versus $g_{\perp}/2\pi = -100.978$ kHz (out-of-plane tilt, $G_{\perp}$); here $g_{\|}/g_{\perp} = 0.0445 = \mathrm{Re}\, G_{\|}/\,\mathrm{Re}\, G_{\perp}$, a factor of 22 that the single-mode Hamiltonian cannot express.
3. **Exact normal modes.** $\omega_{\pm} = 1.442342$ and 1.239810 MHz (each doubly degenerate), matching $\omega_t\sqrt{1 \pm 2g/\omega_t}$ to 4×10^{-9} rad/s. The RWA form $\omega_t \pm g$ is off by $g^2/2\omega_t = 3.79$ kHz, which quantifies the error of Eq. (14) itself.
4. **Triaxial particle** with semi-axes $300, 220, 150$ nm: the two librations are non-degenerate, at 1.591 and 1.010 MHz, and carry different couplings, $g_{22}/2\pi = -156.5$ kHz and $g_{33}/2\pi = -49.3$ kHz. Two frequency-resolved beam-splitter channels between the particles, not one.
5. **Equivalent ellipsoid.** A prescribed $\boldsymbol{\alpha} = (9.00, 6.40, 5.00) \times 10^{-31}$ maps to semi-axes $(329.9, 200.4, 143.0)$ nm, which reproduce $\boldsymbol{\alpha}$ to five digits.

5 Multipole corrections, and a caution about the published ones

For a centrosymmetric particle the leading corrections are the $O((a/R)^2)$ and $O((a/R)^4)$ terms of Appendix A of Ref. [1],

$$\frac{g^{\mathrm{exact}}}{g^{\mathrm{dip}}} = 1 + C_1 + C_2 + \dots, \qquad C_1 = \frac{3}{5}\frac{(\epsilon_r - 1)(a^2 - b^2)}{\left[1 + L_{\|}(\epsilon_r - 1)\right]^2 R^2}, \qquad C_2 = \frac{9}{25}\frac{(\epsilon_r - 1)^2(a^2 - b^2)^2}{\left[1 + L_{\|}(\epsilon_r - 1)\right]^4 R^4}. \tag{15}$$

Source (at $R = \lambda$, $a = 300$ nm, $b = 180$ nm, $\epsilon_r = 5.7$)	C_1	C_2
Eq. (15) evaluated here	0.0363	0.00132
Ref. [1], Appendix A text	0.0252	0.0006
Ref. [2], `FIG_2(b).csv` header	0.033	0.012

Table 2: The three published/derived values of the multipole coefficients disagree. All are at the percent level, so none of them — nor their combination — accounts for the constant factor 1.334977 by which the Zenodo curve exceeds Eq. (14) of `../challenge20.pdf`.

For a shape *without* inversion symmetry the expansion also contains the odd term generated by A in Eq. (2),

$$\delta U_{\mathrm{int}}^{(a/R)} = -\tfrac{C}{3}\,\mathrm{Re}\big[(\mathsf{A}_1 \cdot \mathbf{E}_0) : \nabla \mathsf{G}(\mathbf{R}) \cdot \mathbf{p}_2\big] + (1 \leftrightarrow 2), \tag{16}$$

which is linear in a/R and therefore *dominates* C_1 for a strongly asymmetric particle at close range. This term, and a possible misalignment of the principal axes of $\boldsymbol{\alpha}$ and I, are the only two places where an arbitrary shape cannot be replaced by its equivalent ellipsoid. Both are outside the scope of the challenge's Hamiltonian; `generalized.py` implements the dipole order exactly and Eq. (15) for reference, and does *not* implement A.

References

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